👩‍🏫 Staff Guide · Small Engines Overhaul · Unit 6 Ignition (Level 1)

Level 1 = about 2nd-grade reading, with emoji and pictures. SLE students stay in the class teacher's lesson and use these pieces next to it: same topics, same order, same points.

1. How the pieces line up

Gen ed piece (Canvas: “Week 4: Ignition Handout, iCADEMY Bronze 7 Magneto Ignition Operation”)	Level 1 piece
Ignition Handout (title only)	Deck + notes + quiz (team-written)
iCADEMY Bronze 7 Magneto Ignition	Our own slides 'The Magneto' (no STIHL content)
Class slides	Deck: 13 slides, Words to Know first, “point to the picture” checks, recap
His worksheets (grouped into lessons)	Guided notes: 15 items + key
Class quiz / notes check	Quiz: 8 questions + key
Vocabulary	8 picture cards (cut apart)
—	1 game on the class site: 10 questions + a matching round

2. Supports built in

- **Read aloud** every line; student may point, say, or circle (point = full answer)
- One idea per slide; big pictures; 2–3 picture choices
- Show a real flywheel and armature side by side
- Spark plug passed around (clean, cold)

3. Add these as the IEP allows

- Read-aloud · extended time · separate setting
- Printed slides · videos paused at the “look for” part
- Cover one choice to make it 2

4. What to grade

- Ignition Handout: use the notes as the handout; same points
- **Notes**: same points as his worksheets for that lesson when complete.
- **Quiz**: score, then scale to the class points.
- These are **modifications** (less content). The IEP must allow modified grading. Check with the case manager.

5. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm in his files say “check with the class teacher”.

🚧 Under construction (check with the class teacher)

- **Ignition Handout**: Title only in Canvas; our notes and quiz are team-written.

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
- **Eyes and body**: safety glasses on in the shop at all times; coveralls, closed-toe shoes, no loose clothing, long hair tied back, no dangling jewelry.
- **Fuel and oil**: gasoline is flammable. Drain fuel with adult supervision, outside or in a ventilated area, no sparks or flames nearby. Oily rags go in the metal safety can. Gloves for oil; wash hands after.
- **Exhaust**: engines are only started with adult supervision in a well-ventilated area or outdoors (carbon monoxide). Hot mufflers and cylinders burn: let parts cool.
- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
- **Before working on an engine**: disconnect the spark plug wire so the engine cannot start.
- **Lifting**: engines are heavy; lift with a partner, bend the knees. Car lifts, jacks and 18-wheeler work are video topics in this class.
- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.

👩🏾‍🏫 Staff Guide · Spark Plug & Spark Check (Level 1)

Small Engines Overhaul · Unit 6 · goes with the class “Ignition Handout” (lab points). Level 1 = about 2nd-grade reading. Same activity, same day, same points as the class.

1. The pieces

- **Task card**: Modified from the class “Ignition Handout”. 7 steps, one per box.
- **Prep sheet**: jobs, tools checklist, safety. Do it the day before or at the start.
- **Slides**: jobs, safety, steps, 3 checks.
- **Reflection**: replaces the written parts.
- **Game**: on the class site.

2. Prompts while they work

- Show me the first step.
- Which tool do we need now?
- Where do your hands go while the tool is turning?
- What should we do when we finish?

3. Modifications

- Model first, then the student tries with adult supervision
- Picture task card at the bench
- Parts tray with labeled cups for small parts

4. What to grade (lab points)

- Did the steps with support (photo proof)
- My Check page done
- Pointing, saying, or circling counts as a full answer.
- Modified grading only as the IEP allows. Check with the case manager.

5. Ask

- Mr. Driver (SLE).

🚧 Under construction (check with the class teacher)

- **This packet**: Plug and air gap specs depend on the engines in Mr. Davis's shop; staff fill in the spec.

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
- **Eyes and body**: safety glasses on in the shop at all times; coveralls, closed-toe shoes, no loose clothing, long hair tied back, no dangling jewelry.
- **Fuel and oil**: gasoline is flammable. Drain fuel with adult supervision, outside or in a ventilated area, no sparks or flames nearby. Oily rags go in the metal safety can. Gloves for oil; wash hands after.
- **Exhaust**: engines are only started with adult supervision in a well-ventilated area or outdoors (carbon monoxide). Hot mufflers and cylinders burn: let parts cool.
- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
- **Before working on an engine**: disconnect the spark plug wire so the engine cannot start.
- **Lifting**: engines are heavy; lift with a partner, bend the knees. Car lifts, jacks and 18-wheeler work are video topics in this class.
- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.

⚡ Ignition Notes

Small Engines Overhaul · Unit 6 · Level 1

Name: _____

✏️ A. Trace the word. Then copy it.

⚡ spark _____

🧲 magnet _____

🌀 coil _____

📏 gap _____

🖼️ B. Fill in the word. Use the pictures.

5. ⚡ The plug makes a _____.

6. 🧲 The flywheel has a _____.

7. 🌀 The _____ makes electricity.

8. 📏 A _____ is a small space.

Word bank: 🌀 **coil** | 📏 **gap** | 🧲 **magnet** | ⚡ **spark**

⭕ C. Circle the right picture.

9. Do small engines need a battery for spark?



Yes, always



No, magnets

10. A good spark is…



None



Blue

11. The kill switch…



Stops the engine



Speeds it up

🔗 D. Draw a line from the picture to the word.









magnet

coil

tester

spark plug

📖 Ignition: Vocabulary Picture Cards (Level 1)

Name: _____

Cut on the dashed lines. Use the cards to match, sort, or quiz a partner.



Spark plug

makes the spark



Magneto

magnets that make electricity



Armature (coil)

the coil near the flywheel



Air gap

space between coil and flywheel



Spark plug gap

space at the plug tip



Flywheel key

small key that lines up the flywheel



Kill switch

stops the engine



Spark tester

tool to see the spark

⚡ Spark Plug & Spark Check

Small Engines Overhaul · Unit 6 · Level 1



Name: _____

Modified from the class “Ignition Handout”.

🧺 Supplies: get it. Check ☐.

- ☐ 🔧 **Spark plug socket** + ratchet
- ☐ 📏 **Gap gauge**
- ☐ 🧪 **Spark tester**
- ☐ 📄 **Gauge card** air gap
- ☐ ⚡ **New spark plug** optional
- ☐ 📋 **Plug color chart**

⚡ Do one box. Check ☐ it.

1		Glasses on.	☐
2		Plug wire off.	☐
3		Take out the plug with an adult.	☐
4		Look at the color.	☐
5		Check the gap with an adult.	☐
6		Spark tester on.	☐
7		Pull. See a blue spark?	☐

🚧 **Under construction:** Plug and air gap specs depend on the engines in Mr. Davis's shop; staff fill in the spec. *Check with the class teacher.*

⚡ Spark Plug & Spark Check · Get Ready

Small Engines Overhaul · Unit 6 · Level 1

Name: _____

👥 My job. Circle it.

- **Mechanic** — fixes
- **Tool helper** — gets tools
- **Recorder** — writes
- **Clean-up captain** — cleans

🧰 Get the tools. Check ☐.

- ☐ Spark plug socket
- ☐ Gap gauge
- ☐ Spark tester
- ☐ Gauge card
- ☐ New spark plug
- ☐ Plug color chart

⚠️ Be safe!

- Glasses on.
- Coveralls on.
- Tools with an adult.
- Use the tester.
- Clean up, wash hands.

✅ Before we start

☐ Safety glasses ☐ Coveralls ☐ Adult at bench

⚡ Spark Plug & Spark Check · My Check

Small Engines Overhaul · Unit 6 · Level 1



Name: _____

📸 My photos. Check ☐.

- ☐ 📸 Before
- ☐ 📸 During
- ☐ 📸 After

✅ I did it! Check ☐.

- ☐ 🥽 I wore my safety gear
- ☐ 🔧 I did the engine job
- ☐ 🧽 I cleaned up
- ☐ 🤝 I helped my group

⭕ Circle one face.

🥽 **I was safe.**	😀	🙂	😕
🔧 **I did my job.**	😀	🙂	😕
⭐ **I liked it.**	😀	🙂	😕

⭐ I learned ____. Draw or write.

✅ Ignition Quiz

Small Engines Overhaul · Unit 6 · Level 1 · 8 questions. Circle the picture.

Name: _____

1. ⚡ Makes the spark…

-  Muffler
-  Spark plug

2. 🧲 Flywheel has…

-  Magnets
-  Cookies

3. 🌀 Makes electricity…

-  Coil
-  Sponge

4. 📏 Space at plug tip…

-  Hole in tank
-  Gap

5. 🔌 Stops the engine…

-  Yelling
-  Kill switch

6. 🧪 See the spark…

-  Spark tester
-  Lick it

7. 🔵 Good spark color…

-  None
-  Blue

8. 🔋 Small engine spark needs…

-  Magnets
-  Big battery

⚡ Ignition Notes · KEY

Small Engines Overhaul · Unit 6 · Level 1

Name: _____

✏️ A. Trace the word. Then copy it.

⚡ spark _____

🧲 magnet _____

🌀 coil _____

📏 gap _____

🖼️ B. Fill in the word. Use the pictures.

5. ⚡ The plug makes a **spark**.

6. 🧲 The flywheel has a **magnet**.

7. 🌀 The **coil** makes electricity.

8. 📏 A **gap** is a small space.

Word bank: 🌀 coil | 📏 gap | 🧲 magnet | ⚡ spark

⭕ C. Circle the right picture.

9. Do small engines need a battery for spark?

-  Yes, always
-  No, magnets ✓

10. A good spark is…

-  None
-  Blue ✓

11. The kill switch…

-  Stops the engine ✓
-  Speeds it up

🔗 D. Draw a line from the picture to the word.

Key: ⚡ → spark plug · 🧲 → magnet · 🌀 → coil · 🧪 → tester

⚡	magnet
🧲	coil
🌀	tester
🧪	spark plug

🔑 *Key written by our team from the class teacher's materials, please check.*

✅ Ignition Quiz · KEY

Small Engines Overhaul · Unit 6 · Level 1 · 8 questions. Circle the picture.

Name: _____

1. ⚡ Makes the spark… Ignition Handout

- 🔇 Muffler
- ⚡ Spark plug **(correct)**

2. 🧲 Flywheel has… Ignition Handout

- 🧲 Magnets **(correct)**
- 🍪 Cookies

3. 🌀 Makes electricity… Ignition Handout

- 🌀 Coil **(correct)**
- 🧽 Sponge

4. 📏 Space at plug tip… Ignition Handout

- 🕳️ Hole in tank
- 📏 Gap **(correct)**

5. 🔌 Stops the engine… Ignition Handout

- 📣 Yelling
- 🔌 Kill switch **(correct)**

6. 🧪 See the spark… Notes

- [x] 🧪 Spark tester
- [ ] 👅 Lick it

7. 🔵 Good spark color… Notes

- [ ] ⚫ None
- [x] 🔵 Blue

8. 🔋 Small engine spark needs… Notes

- [x] 🧲 Magnets
- [ ] 🔋 Big battery

Answers

1-B (Spark plug) · 2-A (Magnets) · 3-A (Coil) · 4-B (Gap) · 5-B (Kill switch) · 6-A (Spark tester) · 7-B (Blue) · 8-A (Magnets)

Level 1 version of the gen ed “Ignition Handout (team-written)”. Read every question aloud. Score: __ / 8.

🔑 *Key written by our team from the class teacher's materials, please check.*